Artificial Intelligence & Machine Learning Internship

The internship report submitted in partial fulfilment of the requirements for the award of degree

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

C GURUTEJA

22191A0507

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

JNTUA COLLEGE OF ENGINEERING (Autonomous), PULIVENDULA

ANDHRA PRADESH - INDIA

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Internship report on **Artificial Intelligence & Machine Learning Internship.** A record of internship work done and submitted by C GURUTEJA bearing Roll No.22191A0507.

Signature of Guide

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ACKNOWLEDGEMENT

It is a great pleasure to express a deep sense of gratitude and veneration to the internship provider **CODEC Technologies** for giving me an internship opportunity.

I express my sincere thanks to **Dr. D. Vishnu Vardhan**, Principal of JNTUA College of Engineering (Autonomous), Pulivendula, for his encouragement and for providing the opportunity for the students to go and work in the field with the advent of internship while pursuing B.Tech IVth year.

I express my sincere thanks to **Prof. K. Sesha Maheswaramma**, Vice-Principal of JNTUA College of Engineering (Autonomous), Pulivendula, for her encouragement and suggestions.

Foremost, I gratefully express my sincere gratitude to our respected **Dr. S. Jessica Saritha**, Associate Professor, Head of the Department, Department of Computer Science and Engineering, JNTUA College of Engineering, Pulivendula, for his valuable guidance, timely suggestions, and lively encouragement throughout the work. I am always thankful to him for providing me with the opportunity to do this work.

I am especially thankful to my guide **Dr. K. Balachandra,** Assistant Professor(A), Department of Computer Science and Engineering, JNTUA College of Engineering, Pulivendula, for his valuable guidance, constant encouragement, and support during the internship. His mentorship has been instrumental in successfully completing my project on **Artificial Intelligence & Machine Learning** . Her wide knowledge, practical insights, and motivation have inspired me throughout the course of this internship.

With gratitude,

C.GURUTEJA

(22191A0507)

Internship Weekly Report

Week-1

(From 19/01/2026 To 25/01/2026)

Objective of the Activity Done:

To build a strong foundation in Artificial Intelligence and Machine Learning concepts, set up the development environment, and start preliminary work on the "**AI Review Intelligence**" project. Additionally, to gain practical experience in data preprocessing, exploratory data analysis, and understanding model selection strategies for medical diagnosis applications.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Attended internship orientation and introduction session. Understood internship structure, project objectives, and deliverables.	Learned about the AI/ML project flow, the significance of medical data analysis, and overview of CODEC Technologies .
Day 2	Revised basics of Python programming and explored libraries like NumPy, Pandas, and Matplotlib for data handling and visualization.	Strengthened Python coding skills and learned to manipulate and visualize datasets efficiently.
Day 3	Attended live session on Machine Learning concepts — supervised vs. unsupervised learning, regression, and classification.	Understood key ML concepts and their applicability to disease prediction tasks such as AI Review Intelligence-with-Machine-Learning.

Day 4	Collected and analyzed the Reviews data. Performed data cleaning and handled missing values.	Learned data preprocessing techniques and understood the importance of data quality in model performance.
Day 5	Conducted exploratory data analysis (EDA) to identify patterns and correlations between features. Visualized distributions using Matplotlib and Seaborn.	Gained skills in EDA, correlation analysis, and data visualization for better feature understanding.
Day 6	Participated in a Q&A session with mentors to clarify doubts regarding dataset preparation and model approach.	Understood how to design the ML pipeline, manage GitHub version control.

Internship Weekly Report

Week-2

(From 26/01/2026 To 01/02/2026)

Objective of the Activity Done:

To develop the backend logic and machine learning workflow for **AI Review Intelligence** using Jupyter Notebook. The objective was to preprocess the dataset, train the ML model, and evaluate its performance to ensure accurate disease prediction.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Began model development in Jupyter Notebook. Imported dataset and explored its structure.	Learned to load and analyze datasets using Pandas and NumPy.
Day 2	Performed data cleaning, handled missing values, and encoded categorical data.	Understood preprocessing techniques and data transformation for ML readiness.
Day 3	Split dataset into training and testing sets. Implemented different ML algorithms like Logistic Regression and Random Forest.	Split dataset into training and testing sets. Implemented different ML algorithms like Logistic Regression and Random Forest.
Day 4	Evaluated model performance using accuracy, precision, recall, and confusion matrix.	Evaluated model performance using accuracy, precision, recall, and confusion matrix.
Day 5	Tuned model parameters using GridSearchCV and improved prediction accuracy.	Tuned model parameters using GridSearchCV and improved prediction accuracy.
Day 6	Participated in mentor review session to discuss results and improvement scope.	Participated in mentor review session to discuss results and improvement scope.

Internship Weekly Repost

Week-3

(From 02/02/2026 To 09/02/2026)

Objective of the Activity Done:

To enhance **AI Review Intelligence** model by improving accuracy through feature selection and model optimization, implementing proper data handling, and building an interpretable results dashboard for better analysis.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Reviewed previously trained models and compared their performance metrics.	Reviewed previously trained models and compared their performance metrics.
Day 2	Performed feature selection using correlation analysis and Recursive Feature Elimination (RFE).	Performed feature selection using correlation analysis and Recursive Feature Elimination (RFE).
Day 3	Applied advanced ML algorithms such as XGBoost and SVM for performance comparison.	Applied advanced ML algorithms such as XGBoost and SVM for performance comparison.
Day 4	Implemented model evaluation using ROC curves, AUC score, and confusion matrix visualization.	Implemented model evaluation using ROC curves, AUC score, and confusion matrix visualization.
Day 5	Created visualizations in Jupyter Notebook to display prediction outputs and feature impacts.	Created visualizations in Jupyter Notebook to display prediction outputs and feature impacts.
Day 6	Participated in mentor session to review model results and discuss next steps for deployment.	Participated in mentor session to review model results and discuss next steps for deployment.

Internship Weekly Repost

Week-4

(From 09/02/2026 To 15/02/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Report

Week-5

(From 16/02/2026 To 22/02/2026)

Objective of the Activity Done:

To complete full module integration and testing for the **AI Review Intelligence**, ensuring accurate predictions, clear visual outputs, and seamless data flow from input to result generation.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Set up the final Jupyter Notebook environment and organized all project files and datasets.	Set up the final Jupyter Notebook environment and organized all project files and datasets.
Day 2	Integrated data preprocessing, model training, and evaluation modules into a single notebook.	Integrated data preprocessing, model training, and evaluation modules into a single notebook.
Day 3	Connected the trained model with the user input section to generate live predictions.	Connected the trained model with the user input section to generate live predictions.
Day 4	Visualized prediction results and key performance metrics using Matplotlib and Seaborn.	Visualized prediction results and key performance metrics using Matplotlib and Seaborn.
Day 5	Conducted full testing of the pipeline using multiple datasets and test samples.	Conducted full testing of the pipeline using multiple datasets and test samples.
Day 6	Reviewed mentor feedback, refined documentation, and finalized the project notebook.	Reviewed mentor feedback, refined documentation, and finalized the project notebook.

Internship Weekly Report

Week-6

(From 23/02/2026 To 01/03/2026)

Objective of the Activity Done:

To finalize the **AI Review Intelligence** project by testing, evaluating, documenting, and presenting the completed model, marking the successful completion of the internship.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Performed final integration of all modules and fixed minor issues in the ML pipeline.	Performed final integration of all modules and fixed minor issues in the ML pipeline.
Day 2	Conducted end-to-end testing of data preprocessing, model prediction, and visualization sections.	Conducted end-to-end testing of data preprocessing, model prediction, and visualization sections.
Day 3	Prepared detailed project documentation and summarized model accuracy and findings.	Prepared detailed project documentation and summarized model accuracy and findings.
Day 4	Presented the final project to mentors and participated in feedback session.	Presented the final project to mentors and participated in feedback session.

Internship Weekly Repost

Week-7

(From 02/03/2026 To 08/03/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Repost

Week-8

(From 09/03/2026 To 15/03/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly

Repost Week-9

(From 16/03/2026 To 22/03/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Repost

Week-10

(From 23/03/2026 To 29/03/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Repost

Week-11

(From 30/03/2026 To 05/04/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Repost

Week-12

(From 06/04/2026 To 12/04/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

Internship Weekly Repost

Week-13

(From 13/04/2026 To 19/04/2026)

Objective of the Activity Done:

To implement a user-interactive prediction interface for **AI Review Intelligence**, enabling input handling, prediction generation, and automated result display for an enhanced user experience.

DAY	BRIEF DESCRIPION OF THE DAILY ACTIVITY	LEARNING OUTCOME
Day 1	Designed a data input form in Jupyter Notebook for user test values.	Designed a data input form in Jupyter Notebook for user test values.
Day 2	Integrated the trained ML model with the input form for live prediction.	Integrated the trained ML model with the input form for live prediction.
Day 3	Implemented result visualization using Matplotlib and Seaborn.	Implemented result visualization using Matplotlib and Seaborn.
Day 4	Tested multiple test cases to verify prediction accuracy and stability.	Tested multiple test cases to verify prediction accuracy and stability.
Day 5	Enhanced user interface and automated result summarization for reports.	Enhanced user interface and automated result summarization for reports.
Day 6	Attended mentor Q&A session on visualization improvement and documentation.	Attended mentor Q&A session on visualization improvement and documentation.

CONTENTS

Chapter No.	Contents	Page No.
1.	Introduction	**1**
2.	About the Internship	**2**
3.	Internship Specification	**3-5**
4.	Internship Requirements	**6-8**
5.	Modules Learned	**9-10**
6.	Analysis of Modules Completed	**11-13**
7.	Project Work Undertaken: **AI Review Intelligence**	**14-18**
8.	Conclusion	**19**
9.	Internship Certificate	**20**

1.INTRODUCTION

About The Company

CodeC Technologies, together with its flagship initiative, represents a comprehensive educational technology ecosystem designed to bridge the gap between academic learning and industry requirements. **CodeC Technologies** serves as the parent organization, designing and delivering industry-aligned training and skill development programs, while its internship platform provides a specialized environment for virtual internships and project-based learning. This integrated approach enables students across geographies to gain hands-on experience through real-world projects under the guidance of industry mentors, thereby fostering professional readiness and practical competence.

The mission of **CodeC Technologies** is to build a sustainable talent pipeline by equipping learners with the technical expertise, problem-solving ability, and confidence required to excel in competitive environments. Through structured training modules, workshops, and guided mentorship, **CodeC Technologies** emphasizes experiential learning, ensuring participants acquire not only theoretical knowledge but also the skills to apply it effectively in real-world scenarios.

During my internship with **CodeC Technologies**, I was trained in Full Stack Web Development, with a focus on mastering the complete lifecycle of application development. The program covered a wide range of technologies, including front-end development with HTML, CSS, and JavaScript; dynamic client-side programming with React; server-side logic using Node.js and Express.js; and database management through MongoDB. This holistic learning experience provided me with an in-depth understanding of the MERN stack and its applications in building modern, scalable, and efficient web applications.

This internship experience has not only enhanced my technical foundation but has also strengthened my analytical and professional skills. It has served as a bridge between academic concepts and industry practices, preparing me to contribute effectively in the field of full stack development. The chapters that follow provide a detailed overview of the internship program, the modules learned, and the insights gained through this comprehensive training.

2. ABOUT THE INTERNSHIP

The internship offered by **CodeC Technologies** provides students with an immersive experience designed to bridge the gap between academic learning and industry-ready skills. The program emphasizes hands-on, project-based learning, enabling participants to work on real-world challenges while gaining exposure to emerging technologies and practical workflows.

Interns are introduced to a structured learning environment that combines theoretical understanding with practical application. They get the opportunity to collaborate with industry professionals, receive mentorship, and contribute to live projects, enhancing both technical and soft skills. The program covers various domains including web development, artificial intelligence, machine learning, data analysis, and other in-demand technologies.

One of the key highlights of this internship is the focus on building solutions that mirror actual industry scenarios. Interns can develop, test, and deploy projects while understanding the end-to-end lifecycle of software and AI-driven applications. The initiative encourages problem-solving, innovation, and adaptability, ensuring that participants are not only equipped with technical expertise but also with critical thinking and project management skills.

In addition to technical training, the internship emphasizes professional growth. Participants are guided on effective communication, teamwork, and industry best practices. They also gain insights into how companies operate, project delivery standards, and performance expectations, preparing them for future roles in technology organizations.

The **CodeC Technologies** internship is particularly suited for students who wish to transform academic knowledge into practical experience. By the end of the program, interns emerge with a portfolio of completed projects, enhanced technical competence, and a clearer understanding of industry requirements — making them more confident and competitive in the professional landscape.

Overall, this internship blends learning, experience, and mentorship in a way that fosters growth, creativity, and industry readiness, helping students take a significant step toward their career aspirations.

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3.INTERNSHIP SPECIFICATION

What is an AI/ML Developer

An **AI/ML (Artificial Intelligence/Machine Learning) Developer** is a professional skilled in designing, training, and deploying machine learning models to solve real-world problems using data. They analyze large datasets, identify meaningful patterns, and build predictive systems that support data-driven decision-making. AI/ML Developers work across the entire data pipeline — from data collection and preprocessing to model training, evaluation, and deployment. Their goal is to develop intelligent systems that can learn from data, make accurate predictions, and automate tasks efficiently.

Python:

Python is the most widely used programming language in AI and ML due to its simplicity and vast ecosystem of libraries such as **NumPy**, **Pandas**, and **Matplotlib**.
It is used for data manipulation, numerical computations, and building predictive models efficiently.

Data Preprocessing:

Before training an ML model, data needs to be cleaned and prepared. Techniques such as **handling missing values**, **encoding categorical variables**, and **feature scaling** ensure data quality and consistency.
Preprocessing is crucial for improving model accuracy and performance.

Machine Learning Algorithms:

Machine learning algorithms such as **Logistic Regression**, **Random Forest**, **Support Vector Machine (SVM)**, and **XGBoost** are applied to identify disease presence based on medical parameters.
These algorithms learn from training data and make predictions on new patient records for TrafficTelligence-Advanced-Traffic-Volume-Estimation-with-Machine-Learning.

Model Evaluation:

Model performance is evaluated using metrics like **Accuracy**, **Precision**, **Recall**, **F1-Score**, and **ROC-AUC**.

These metrics help in assessing the reliability of the model and its effectiveness in identifying positive and negative cases accurately.

Visualization:

Libraries such as **Matplotlib** and **Seaborn** are used to create visual representations of data distributions, correlations, and prediction outcomes. Visualization helps in understanding feature importance and improving model interpretability for medical use cases.

Role:

✓ **AI/ML Developers** build intelligent systems capable of learning from data and making accurate predictions.

✓ In the **AI Review Intelligence Project**, the role involves cleaning and analyzing medical data, training predictive models, and interpreting outcomes to assist early diagnosis.

✓ They ensure that the developed system is accurate, interpretable, and useful for healthcare professionals.

Tools and Technologies Used:

1. Python:
The primary programming language used for data analysis, model building, and visualization. Its simplicity and extensive libraries make it ideal for AI/ML development.

2. Jupyter Notebook:
An interactive environment used to write, test, and document Python code. It enables step-by-step model development and visualization in a single workspace.

3. NumPy:
A fundamental library for numerical computing, used for handling arrays, mathematical operations, and efficient data processing.

4. Pandas:
Used for data manipulation and analysis. Pandas provides powerful data structures like DataFrames for managing large datasets effectively.

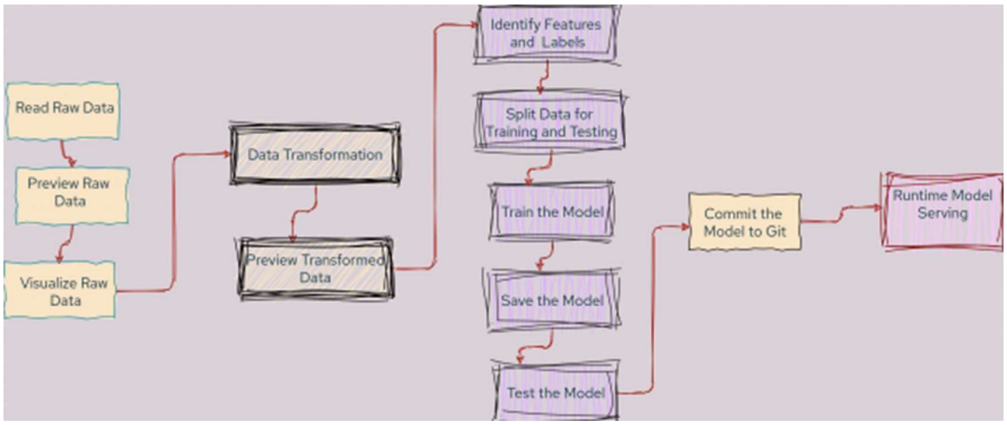
5. Matplotlib & Seaborn:
Visualization libraries used to plot graphs, correlations, and prediction outcomes. They help in understanding data patterns and model performance.

6. Scikit-learn:
A machine learning library that provides tools for classification, regression, model training, and evaluation. It was used to build and test algorithms for liver disease prediction.

7. XGBoost:
An advanced boosting algorithm used to improve model accuracy and performance, especially for classification problems.

8. Google Colab / Local Jupyter Environment:
Used for executing code efficiently with cloud-based GPU support (in Colab) and maintaining notebooks locally for documentation and testing.

9. CSV Files (Dataset):
The dataset containing patient information and liver-related parameters was imported in CSV format for preprocessing and model training.



4.INTERNSHIP REQUIREMNETS

AI/ML Development Environment Setup:

What is AI/ML Development?

Artificial Intelligence (AI) and **Machine Learning (ML)** development involve using data-driven algorithms and statistical models to enable computers to learn from data and make predictions or decisions without being explicitly programmed. In this internship, the **AI Review Intelligence Project** focuses on using ML techniques to analyze the traffic estimation in the populated areas around your sorroundings.

AI/ML developers use tools like **Python, Jupyter Notebook, Scikit-learn, Pandas, and NumPy** to build, train, and evaluate predictive models efficiently.

Step 1: Install Python (Programming Language)

1. Visit the official Python website: https://www.python.org/downloads/
2. Download the latest stable version (preferably Python 3.9 or above).
3. During installation, check the box **"Add Python to PATH"**.
4. Verify installation by opening a command prompt (or terminal) and running:
5. python --version
6. pip –version

These commands confirm that Python and pip (Python package manager) are installed correctly.

Step 2**: Install Jupyter Notebook / Google Colab**

Option 1 – Local Jupyter Notebook:

1. Open Command Prompt or Anaconda Prompt.
2. Install Jupyter using:
3. pip install notebook
4. Launch Jupyter:
5. jupyter notebook
6. A browser window will open where you can create and run .ipynb notebooks.

Option 2 – Google Colab:
Alternatively, visit https://colab.research.google.com/ to use an online notebook environment with built-in GPU support.

Step 3**: Install Required Python Libraries**

Use the following commands to install the essential ML libraries: pip

install numpy

pip install pandas
pip install matplotlib
pip install seaborn
pip install scikit-learn
pip install xgboost

These libraries are used for numerical operations, data analysis, visualization, and model building.

Step 4: **Dataset Setup**

1. Obtain the **AI Review Intelligence** (in CSV format) from verified sources such as Kaggle or UCI Machine Learning Repository.
2. Place the dataset in the project directory for easy access.
3. Load it in Jupyter using Pandas:
4. import pandas as pd
5. data = pd.read_csv('liver_cirrhosis.csv')

Step 5: **Environment Verification**

1. Open Jupyter Notebook.
2. Import all required libraries:
3. import numpy as np
4. import pandas as pd
5. import matplotlib.pyplot as plt
6. import seaborn as sns
7. from sklearn.model_selection import train_test_split
8. from sklearn.ensemble import RandomForestClassifier
9. If no errors appear, your environment is ready for ML development.

Step 6: **Version Control and Project Backup**

1. Install **Git** from https://git-scm.com/.

2. Configure user details:
3. git config --global user.name "Your Name"
4. git config --global user.email "your.email@example.com"
5. Create a GitHub repository to back up code and track progress:
6. git init
7. git add .
8. git commit -m "Initial commit - **AI Review Intelligence** "
9. git remote add origin https://github.com/yourusername/liver-cirrhosis-detection.git
10. git push -u origin main

Step 7**: Optional Tools for AI/ML Development**

- **Anaconda Navigator** – For managing Python environments and packages easily.
- **Kaggle Notebook** – For running ML experiments online.
- **VS Code / PyCharm** – For script-based ML development.
- **Excel / Google Sheets** – For initial data exploration and basic analysis.

Step 8**: Run and Test the Project**

1. Open the Jupyter Notebook containing your ML workflow.
2. Execute each cell sequentially — from **data preprocessing** to **model training** and **evaluation**.
3. Verify output graphs, accuracy results, and predictions.
4. Save your notebook as a final version for documentation and presentation.

5.MODULES LEARNED

Overview of AI/ML Development

The AI/ML internship for **AI Review Intelligence** provides participants with a comprehensive understanding of building predictive models and data-driven applications. This internship offers hands-on experience in **Python-based AI/ML development**, including data preprocessing, exploratory data analysis (EDA), machine learning model training, evaluation, and result visualization. Participants gain proficiency in tools like **Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn**, which are essential for analyzing medical datasets and developing predictive systems. The internship serves as a stepping stone for individuals aspiring to pursue careers in AI/ML, healthcare analytics, and data science.

Purpose

The purpose of the AI/ML internship is to provide participants with structured exposure to the entire ML workflow — from understanding medical datasets to developing models for liver disease prediction. Through step-by-step modules, hands-on exercises, and mentorship, interns gain the knowledge and practical skills required to analyze datasets, train models, interpret results, and present predictions effectively. The internship prepares participants to handle real-world AI/ML challenges while refining their problem-solving, analytical, and professional skills.

Objectives

The objectives of the AI/ML internship include:

1. **Skill Development:** Gain practical experience in AI/ML development using Python and its libraries.
2. **Understanding ML Workflow:** Learn the complete ML pipeline, including data preprocessing, model selection, training, evaluation, and visualization.
3. **Module-Based Learning:** Cover structured modules including Python programming, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and XGBoost.
4. **Mentorship:** Receive guidance from experienced data scientists and AI professionals to navigate complex ML concepts.
5. **Collaborative Learning:** Work in a team environment to enhance knowledge sharing and peer learning.
6. **Professional Development:** Build essential skills such as analytical thinking, problem-solving, and time management.

Need of the Modules

The internship modules address the following needs of participants:

1. **Hands-on Experience:** Apply theoretical AI/ML knowledge to real-world medical datasets.
2. **Guidance and Mentorship:** Receive expert support for complex topics like feature engineering, model selection, and hyperparameter tuning.
3. **Structured Learning:** Progress systematically through modules covering data preprocessing, model building, and evaluation.
4. **Feedback and Evaluation:** Regular assessments to track learning and improve coding and modeling skills.
5. **Networking Opportunities:** Collaborate with peers and mentors to explore professional opportunities.
6. **Career Preparation:** Develop skills required for junior AI/ML roles and healthcare analytics positions.

Characteristics of the Modules

- **Comprehensive Curriculum:** Covers Python, data manipulation, visualization, ML algorithms, and model evaluation.
- **Hands-on Exercises:** Includes practical tasks like dataset preprocessing, EDA, ML model training, hyperparameter tuning, and visualization.
- **Expert Mentorship:** Guidance from professionals to clarify advanced topics and best practices in AI/ML.
- **Collaborative Learning Environment:** Encourages teamwork, discussion, and knowledge sharing.
- **Structured Training:** Modules organized with clear milestones for each phase of the ML workflow.
- **Feedback and Evaluation:** Offers constructive reviews to help interns refine modeling and analytical skills.
- **Career Development Support:** Exposure to real-world tools like Jupyter Notebook, Git, and cloud-based ML environments.

Benefits of the Modules

- **Hands-on Learning:** Practical experience with Python and ML libraries for real-world healthcare applications.
- **Skill Development:** Gain in-demand skills in AI/ML, enhancing employability in data science and healthcare analytics.
- **Expert Guidance:** Mentorship for tackling challenging concepts like feature selection, model evaluation, and visualization.
- **Networking Opportunities:** Build connections with peers and industry professionals.

6.ANALYSIS OF THE MODULES COMPLETED AS A PART OF INTERNSHIP

The AI/ML internship for **AI Review Intelligence** provided a structured learning journey through a comprehensive curriculum designed to equip interns with the skills to build predictive models and analyze Traffic datasets. The modules progressed systematically from foundational Python programming and data handling to advanced machine learning model training, evaluation, and visualization, ensuring a thorough understanding of AI/ML development. Below is a detailed analysis of each module, including content, practical applications, challenges, and contributions to building expertise in AI/ML for healthcare analytics.

1. Python Programming & Data Handling (Sessions 1–4)

- **Content Covered:** Python basics, NumPy, Pandas, data structures, functions, and data manipulation techniques.
- **Practical Applications:** Loaded datasets, performed data cleaning, handled missing values, and transformed features for model readiness.
- **Challenges:** Understanding Pandas DataFrames, handling NaN values, and efficiently manipulating large datasets.
- **Analysis:** These sessions laid the foundation for all AI/ML workflows. Hands-on exercises allowed interns to manage data effectively, ensuring it was clean and structured for model training.

2. Exploratory Data Analysis (EDA) & Visualization (Sessions 5–8)

- **Content Covered:** Data visualization using Matplotlib and Seaborn, correlation analysis, feature distributions, and identifying patterns in medical data.
- **Practical Applications:** Visualized patient health parameters, identified key features related to liver cirrhosis, and explored dataset trends.
- **Challenges:** Choosing appropriate visualization techniques and interpreting medical data correlations.
- **Analysis:** These modules enhanced interns' ability to extract meaningful insights from datasets, crucial for understanding factors influencing disease prediction.

3. Machine Learning Algorithms (Sessions 9–14)

- **Content Covered:** Supervised learning concepts, classification algorithms (Logistic Regression, Random Forest, SVM, XGBoost), model training, and hyperparameter tuning.

- **Practical Applications:** Trained models on Traffic estimation in the surroundings of populated areas compared multiple algorithms, and optimized performance using hyperparameter tuning.
- **Challenges:** Selecting suitable algorithms, avoiding overfitting, and balancing accuracy with interpretability.
- **Analysis:** These sessions developed practical skills in implementing AI/ML models, selecting appropriate algorithms, and understanding their strengths and limitations in medical predictions.

4. Model Evaluation & Metrics (Sessions 15–18)

- **Content Covered:** Accuracy, precision, recall, F1-score, confusion matrix, ROC curves, and AUC evaluation.
- **Practical Applications:** Evaluated the performance of trained models, compared results, and selected the best-performing model for deployment.
- **Challenges:** Understanding metric significance in medical diagnosis scenarios and ensuring models are reliable for healthcare decisions.
- **Analysis:** Interns gained critical skills in evaluating ML models rigorously, ensuring predictions are accurate and clinically meaningful.

5. Model Deployment & User Interaction (Sessions 19–22)

- **Content Covered:** Integrating ML models with interactive input forms in Jupyter Notebook, generating live predictions, and visualizing outcomes.
- **Practical Applications:** Built an interface for users to input patient parameters and obtain cirrhosis predictions with graphical results.
- **Challenges:** Connecting user inputs to model predictions, handling exceptions, and visualizing results effectively.
- **Analysis:** These modules reinforced end-to-end ML pipeline understanding, bridging data preprocessing, model prediction, and result visualization.

6. Project Testing, Validation & Documentation (Sessions 23–26)

- **Content Covered:** Testing models on new data, validating predictions, refining preprocessing, and documenting workflow and results.
- **Practical Applications:** Verified model reliability, prepared detailed project reports, and ensured reproducibility of results.

- **Challenges:** Ensuring data consistency, debugging prediction errors, and maintaining organized documentation.
- **Analysis:** These sessions emphasized the importance of thorough testing, validation, and documentation, preparing interns to present a complete, professional AI/ML project.

Overall Impact

The curriculum's progression from Python fundamentals to advanced ML, evaluation, and visualization provided a holistic AI/ML education tailored to Traffic applications. Hands-on exercises, such as dataset preprocessing, model training, metric evaluation, and interactive prediction visualization, reinforced theoretical concepts and developed practical skills. Exposure to tools like **Jupyter Notebook, Pandas, Matplotlib, Seaborn, Scikit-learn, and XGBoost** aligned the internship with industry standards. Challenges like selecting algorithms, handling traffic data nuances, and connecting prediction workflows fostered analytical thinking, problem-solving, and resilience.

By covering the complete AI/ML workflow, the internship ensured interns could develop end-to-end predictive solutions, preparing them for roles in **AI/ML**

7.PROJECT WORK UNDERTAKEN

Introduction

The **AI Review Intelligence** project is designed to provide an automated, data-driven system for accurate and early estimation of traffic volume based on real-time and historical traffic parameters. The project leverages machine learning techniques to analyze traffic datasets, predict traffic congestion levels, and visualize results in an interactive and interpretable manner.

The brainstorming phase focused on addressing challenges in traffic management, such as manual data analysis, delayed congestion detection, inconsistent traffic interpretation, and the lack of accessible predictive tools for urban traffic planning.

The system is developed using Python and Jupyter Notebook for model development, Pandas and NumPy for data processing, Matplotlib and Seaborn for visualization, and Scikit-learn/XGBoost for predictive modeling. It includes interactive input forms for traffic parameters, automated model-based predictions, and visual dashboards for clear and actionable insights

System Features

Traffic Data Input:

Users or sensors can provide key traffic parameters such as vehicle count, time of day, weather conditions, road type, and average speed.

Predictive Analysis:

Trained machine learning models analyze the given input to predict traffic volume levels and potential congestion trends.

Result Visualization:

The system provides graphical outputs such as traffic flow charts and congestion heatmaps for easy interpretation of predictions.

Model Evaluation:

The performance of the predictive models is validated using metrics like accuracy, precision, recall, F1-score, and R^2/MAE to ensure reliable estimation results **Architecture**

The system follows a **data-driven ML pipeline**:

1. **Data Collection & Preprocessing:** Load CSV dataset, handle missing values, normalize features, and encode categorical variables.
2. **Exploratory Data Analysis (EDA):** Visualize feature distributions, correlations, and trends using Matplotlib/Seaborn.
3. **Model Training:** Train and validate models like Logistic Regression, Random Forest, SVM, and XGBoost.
4. **Prediction Interface:** Collect new patient data, generate predictions, and display visual outcomes.
5. **Evaluation & Documentation:** Test models on unseen data, validate results, and document the workflow for reproducibility.

The architecture ensures end-to-end functionality, from raw data input to actionable predictive output.

Setup Instructions

1. Install **Python 3.9+** and **Jupyter Notebook**.
2. Install required libraries: numpy, pandas, matplotlib, seaborn, scikit-learn, xgboost.
3. Clone the project repository (if applicable) and place the dataset in the project folder.
4. Open the Jupyter Notebook and execute cells sequentially, from **data preprocessing** to **prediction and visualization**.
5. Enter test patient data in interactive forms to generate live predictions.

Running the Application

Patient Workflow:

- Input patient clinical parameters into the system.
- Run the prediction pipeline to receive cirrhosis likelihood.
- View results through graphical representations and summary metrics.

Model Workflow:

- Preprocess new and existing datasets.
- Train or update predictive models with fresh data.
- Evaluate model performance using metrics such as accuracy, F1-score, and ROC-AUC.

Folder Structure

- **Dataset:** Contains patient data in CSV format.
- **Notebooks:** Jupyter Notebook files for preprocessing, training, evaluation, and prediction.
- **Scripts:** Python scripts for utility functions, visualization, and model deployment.

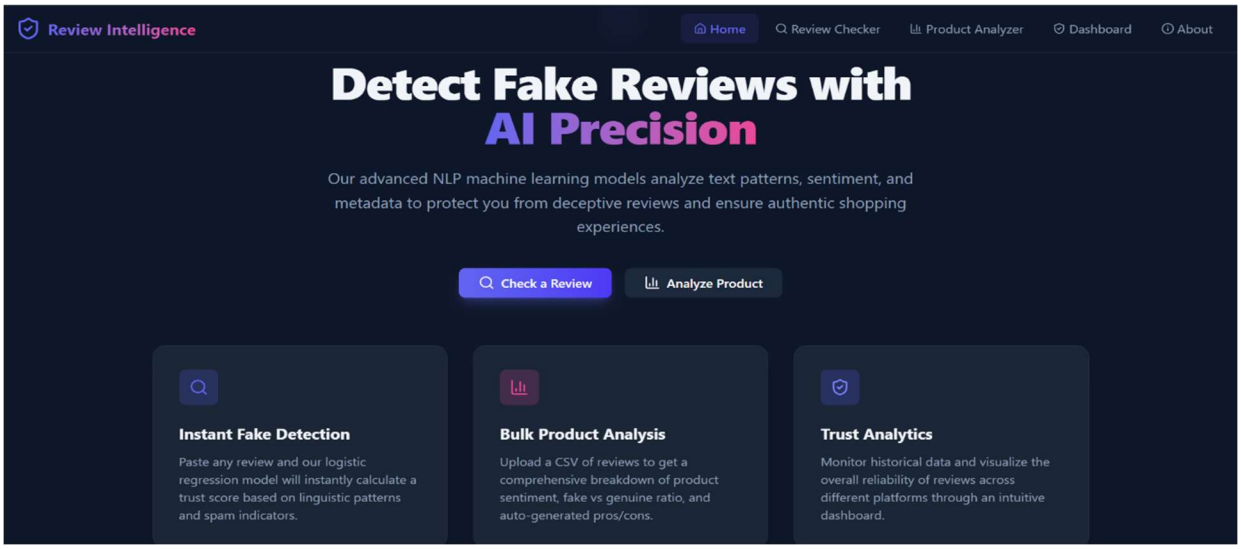
Model Development

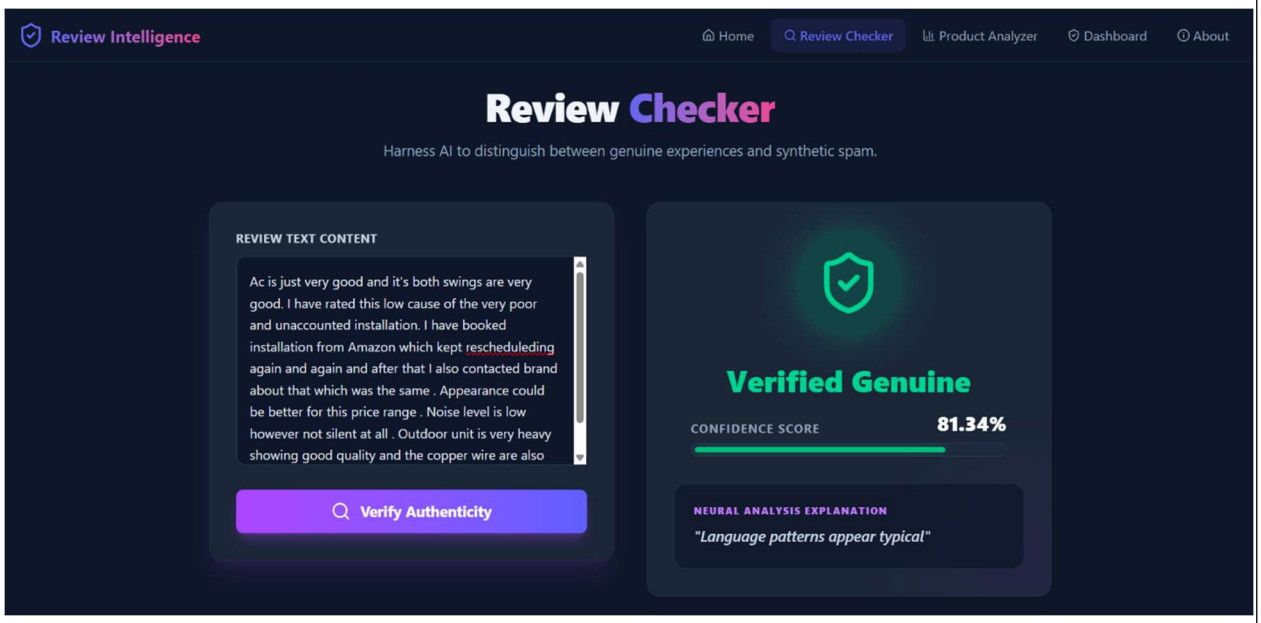
- Preprocessing: Data cleaning, feature scaling, handling missing values.
- Model Training: Supervised learning using Logistic Regression, Random Forest, SVM, XGBoost.
- Evaluation: Accuracy, precision, recall, F1-score, ROC-AUC.
- Prediction Interface: Interactive input for live patient prediction with graphical output.

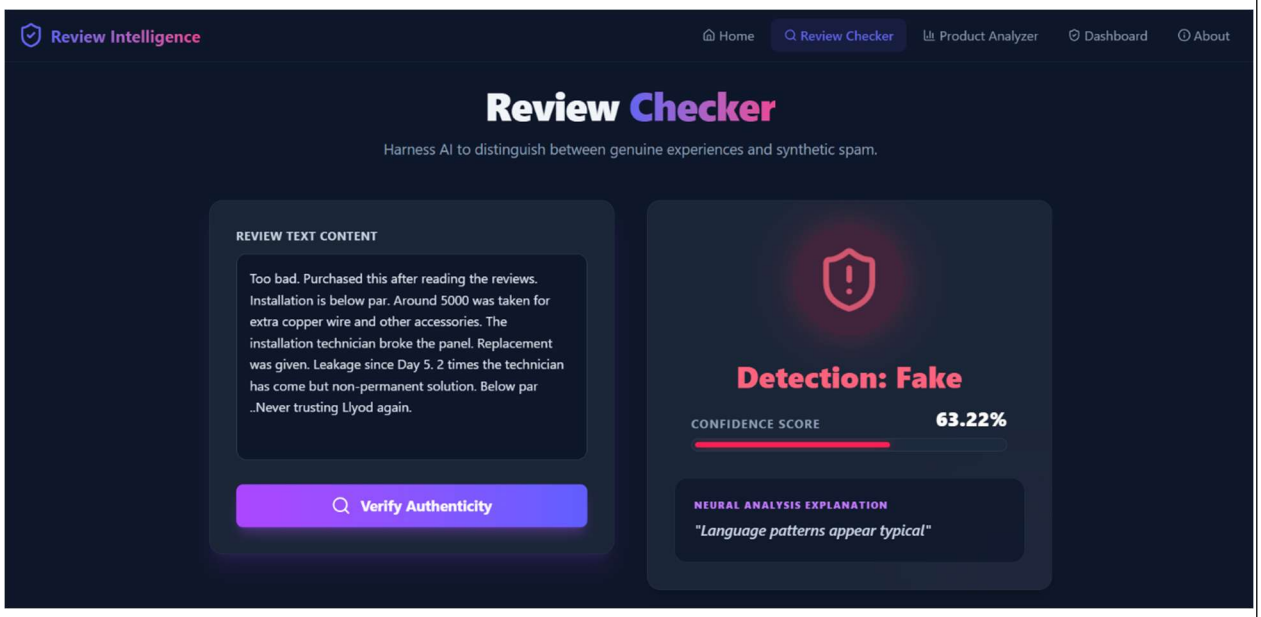
Testing

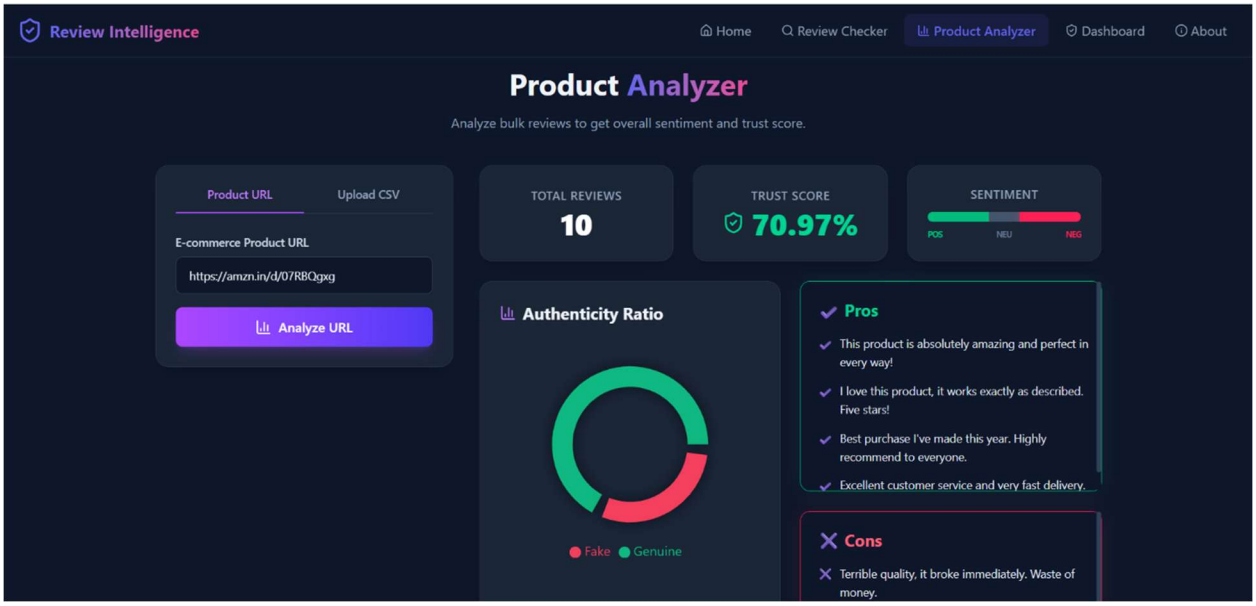
- Tested predictions with multiple patient samples to verify reliability.
- Validated model performance across various evaluation metrics.
- Ensured reproducibility and consistency in prediction outputs.
- Confirmed visualization and interpretation of results were clear and user-friendly.

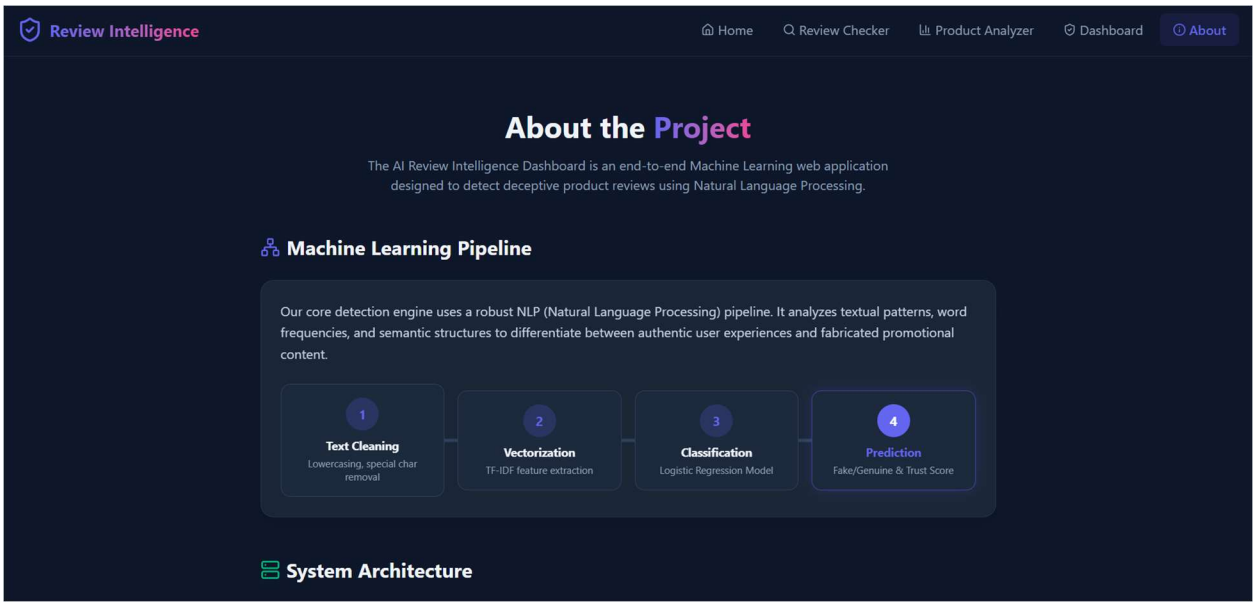
OUTPUTS:

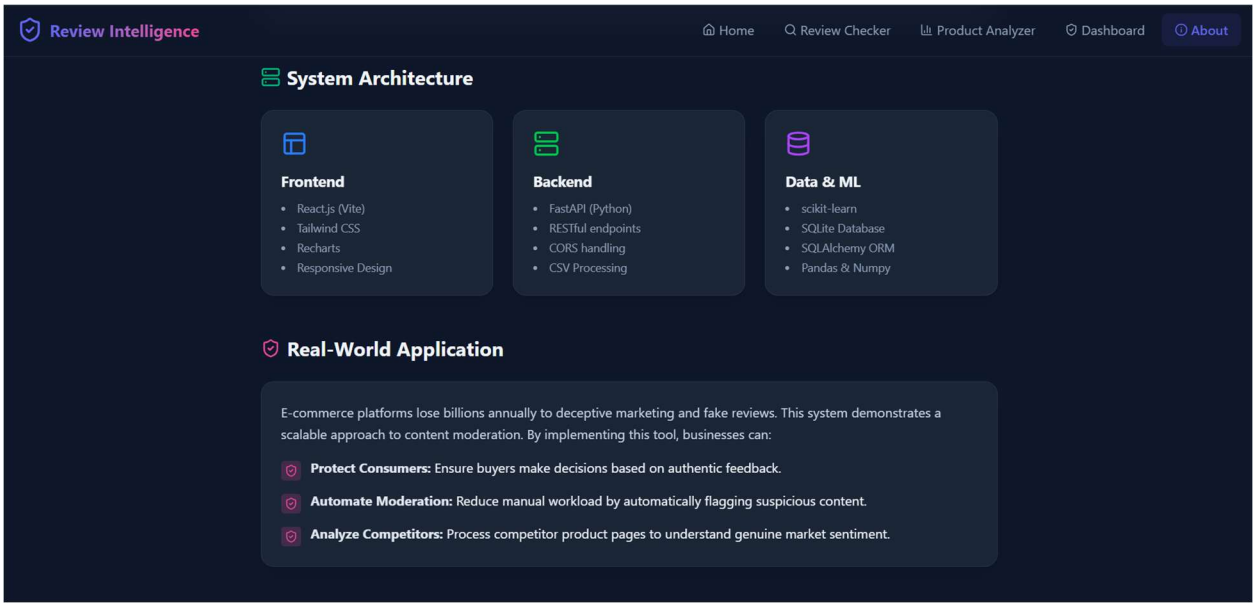

Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
Detect Fake Reviews with AI Precision
Our advanced NLP machine learning models analyze text patterns, sentiment, and metadata to protect you from deceptive reviews and ensure authentic shopping experiences.
Check a Review
Analyze Product
Instant Fake Detection
Paste any review and our logistic regression model will instantly calculate a trust score based on linguistic patterns and spam indicators.
Bulk Product Analysis
Upload a CSV of reviews to get a comprehensive breakdown of product sentiment, fake vs genuine ratio, and auto-generated pros/cons.
Trust Analytics
Monitor historical data and visualize the overall reliability of reviews across different platforms through an intuitive dashboard.


Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
Review Checker
Harness AI to distinguish between genuine experiences and synthetic spam.
REVIEW TEXT CONTENT
Ac is just very good and it's both swings are very good. I have rated this low cause of the very poor and unaccounted installation. I have booked installation from Amazon which kept rescheduleding again and again and after that I also contacted brand about that which was the same . Appearance could be better for this price range . Noise level is low however not silent at all . Outdoor unit is very heavy showing good quality and the copper wire are also
Verify Authenticity
Verified Genuine
CONFIDENCE SCORE
81.34%
NEURAL ANALYSIS EXPLANATION
"Language patterns appear typical"


Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
Review Checker
Harness AI to distinguish between genuine experiences and synthetic spam.
REVIEW TEXT CONTENT
Too bad. Purchased this after reading the reviews. Installation is below par. Around 5000 was taken for extra copper wire and other accessories. The installation technician broke the panel. Replacement was given. Leakage since Day 5. 2 times the technician has come but non-permanent solution. Below par ..Never trusting Llyod again.
Verify Authenticity
Detection: Fake
CONFIDENCE SCORE
63.22%
NEURAL ANALYSIS EXPLANATION
"Language patterns appear typical"


Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
Product Analyzer
Analyze bulk reviews to get overall sentiment and trust score.
Product URL
Upload CSV
E-commerce Product URL
https://amzn.in/d/07RBQgxg
Analyze URL
TOTAL REVIEWS
10
TRUST SCORE
70.97%
SENTIMENT
POS
NEU
NEG
Authenticity Ratio
Fake
Genuine
Pros
This product is absolutely amazing and perfect in every way!
I love this product, it works exactly as described. Five stars!
Best purchase I've made this year. Highly recommend to everyone.
Excellent customer service and very fast delivery.
Cons
Terrible quality, it broke immediately. Waste of money.


Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
About the Project
The AI Review Intelligence Dashboard is an end-to-end Machine Learning web application designed to detect deceptive product reviews using Natural Language Processing.
Machine Learning Pipeline
Our core detection engine uses a robust NLP (Natural Language Processing) pipeline. It analyzes textual patterns, word frequencies, and semantic structures to differentiate between authentic user experiences and fabricated promotional content.
1
Text Cleaning
Lowercasing, special char removal
2
Vectorization
TF-IDF feature extraction
3
Classification
Logistic Regression Model
4
Prediction
Fake/Genuine & Trust Score
System Architecture


Review Intelligence
Home
Review Checker
Product Analyzer
Dashboard
About
System Architecture
Frontend
React.js (Vite)
Tailwind CSS
Recharts
Responsive Design
Backend
FastAPI (Python)
RESTful endpoints
CORS handling
CSV Processing
Data & ML
scikit-learn
SQLite Database
SQLAlchemy ORM
Pandas & Numpy
Real-World Application
E-commerce platforms lose billions annually to deceptive marketing and fake reviews. This system demonstrates a scalable approach to content moderation. By implementing this tool, businesses can:
Protect Consumers: Ensure buyers make decisions based on authentic feedback.
Automate Moderation: Reduce manual workload by automatically flagging suspicious content.
Analyze Competitors: Process competitor product pages to understand genuine market sentiment.

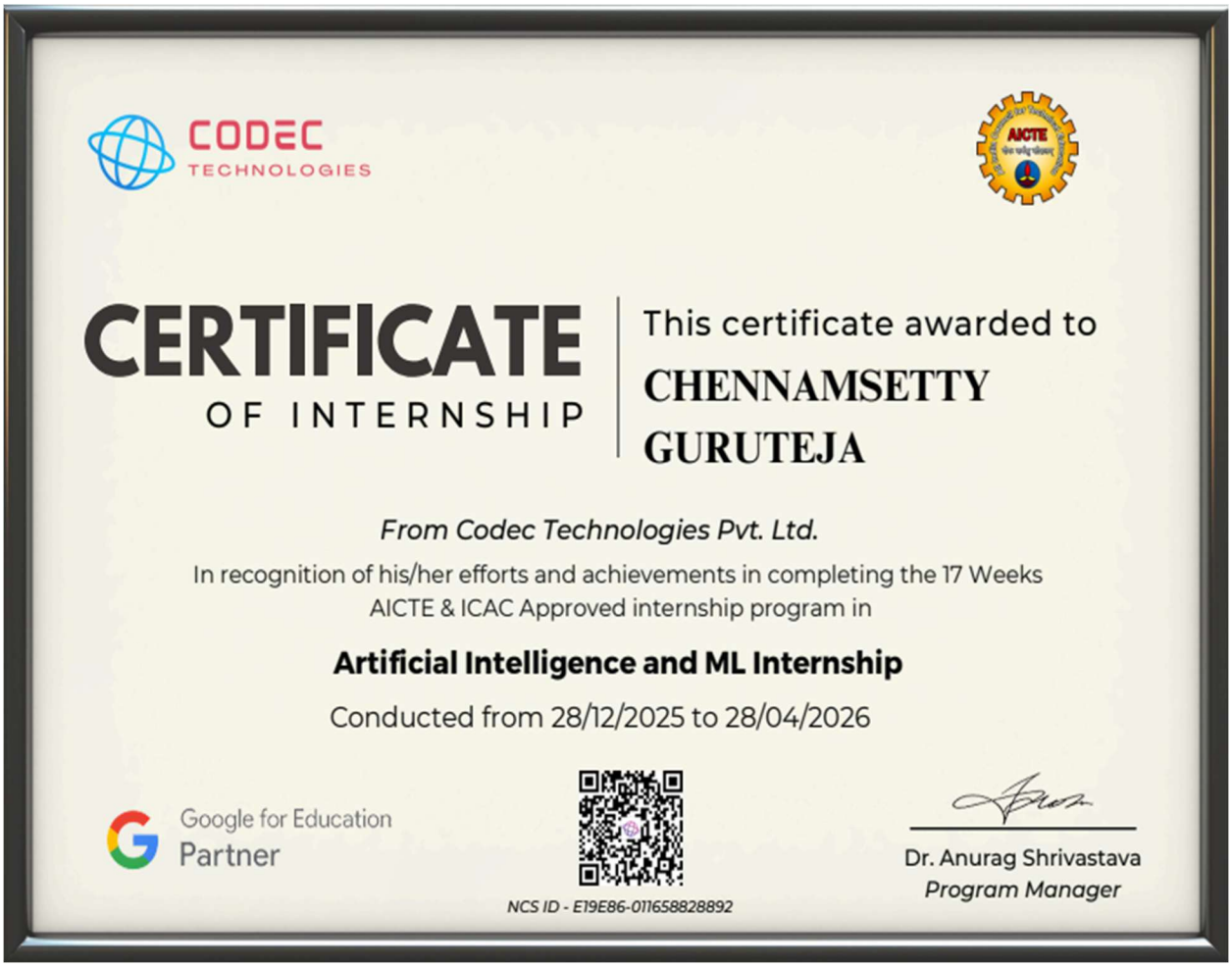
8. CONCLUSION

The AI Review Detection project successfully demonstrates the practical application of Artificial Intelligence and Machine Learning in identifying fake and genuine product reviews. By utilizing Natural Language Processing techniques and classification algorithms, the system effectively analyzes textual data, detects patterns, and provides accurate predictions along with meaningful insights such as sentiment and trust score.

This project helped in gaining hands-on experience in data preprocessing, feature extraction, model training, evaluation, and full-stack integration. It also enhanced understanding of real-world challenges such as handling noisy data, improving model accuracy, and building user-friendly interfaces.

Overall, the AI Review Detection system contributes to improving transparency and reliability in e-commerce platforms by helping users make informed decisions based on trustworthy reviews. The project can be further extended using advanced deep learning models and real-time data integration to achieve even better performance and scalability.

9. INTERNSHIP CERTIFICATE

The internship certificate awarded by **Codec Technologies Pvt. Ltd.** is attached below as proof of successful completion of the **Artificial Intelligence and Machine Learning Internship**.

The certificate confirms that the internship was completed over a duration of **17 weeks (28/12/2025 to 28/04/2026)** and recognizes the efforts and achievements in the field of Artificial Intelligence and Machine Learning.